

Challenge (Habanero)

- Q1.** The vertex of the parabola $y = x^2 + 4x + 4$ is at which point?
 (A) $(0, 0)$
 (B) $(-2, 0)$
 (C) $(-2, 4)$
 (D) $(0, 4)$
 (E) $(-4, 0)$
- Q2.** An online gamer has a score of $s = -2t^2 + 12t + 50$ after t hours, where s is the score and t is the time. What is the axis of symmetry for this parabola?
 (A) $t = 1$
 (B) $t = 2$
 (C) $t = 3$
 (D) $t = 4$
 (E) $t = 5$
- Q3.** A developer wants to optimize the memory usage of a game. The memory usage M (in MB) can be modeled by the equation $M = 3x^2 - 12x + 20$, where x is the number of users. What is the minimum memory usage?
 (A) 2 MB
 (B) 5 MB
 (C) 8 MB
 (D) 10 MB
 (E) 12 MB
- Q4.** For the equation $y = x^2 - 6x + 9$, what are the x -intercepts of the parabola?
 (A) $(-3, 0)$ and $(3, 0)$
 (B) $(-3, 0)$ and $(0, 0)$
 (C) $(0, 0)$ and $(3, 0)$
 (D) $(3, 0)$ only
 (E) $(0, 0)$ only
- Q5.** A data analyst is studying the trend of a new game's popularity. The equation $y = -x^2 + 8x + 12$ models the number of players y over time x . What is the direction of the parabola's opening?
 (A) upward
 (B) downward
 (C) leftward
 (D) rightward
 (E) constant
- Q6.** The equation of a parabola is $y = 2x^2 - 4x - 6$. What is the vertex of this parabola?
 (A) $(-1, -8)$
 (B) $(1, -6)$
 (C) $(-1, 2)$
 (D) $(1, 8)$
 (E) $(-2, 0)$
- Q7.** For the parabola $y = x^2 - 4x + 3$, what is the y -intercept?
 (A) $(-2, 0)$
 (B) $(0, 3)$
 (C) $(2, 0)$
 (D) $(1, 0)$
 (E) $(-1, 0)$
- Q8.** The score s of a gamer in a game is given by $s = -x^2 + 10x - 25$, where x is the level number. What is the maximum score the gamer can achieve?
 (A) 25
 (B) 50
 (C) 75
 (D) 100
 (E) 0

Open Ended Questions (FRQ)

- Q9.** A new app's popularity can be modeled by the equation $y = 2x^2 - 12x + 20$, where y is the number of users and x is the time in days. Graph the parabola, label the vertex, and identify the axis of symmetry. What is the minimum number of users?
- Q10.** The cost C (in dollars) to develop a new game is given by $C = x^2 - 8x + 15$, where x is the number of developers. Find the minimum cost and the number of developers needed to achieve it.

Chef's Tips

- Ensure students have graph paper for questions 9 and 10.
- Remind students to label axes and vertices clearly.
- Encourage students to use calculators to check their answers.